

Andrea Moro^{a,c} – Davide Mocci^{b,d}

^a*Scuola Universitaria Superiore IUSS (Pavia)*; ^c*Scuola Normale Superiore (Pisa)*

^b*University of Cagliari*; ^d*University of Genoa*

3 Cartography and Predication: Why we (still) need exocentricity*

Abstract: Predication has been recognized as the fundamental grammatical relation defining clausal structures in all (and only) human languages ever since the canonical model of Ancient Greek linguistics. This notion is by definition compositional, since either component – the subject and the predicate – cannot be defined without referring to the other. The central question addressed here is whether predication is a lexical property or can instead be unambiguously associated to a dedicated configuration, in keeping with the cartographic guidelines. Capitalizing on copular sentences with predicative noun phrases, which allow us to factor out non-essential aspects of this phenomenon, such as the morphological asymmetry between verbs and nouns, a new approach is illustrated based on the notion of symmetry. This analysis has a deep impact on the cartographic model by reintroducing exocentric phrases into the clausal spine, qualifying the clausal fields and avoiding *ad hoc* solutions that stipulate ‘predicative features’.

Keywords: exocentricity; predicative nominals; predicative feature; subject; symmetry.

3.1 Research question

Predication is an inherently compositional notion: it is the link between a subject and a predicate, neither of which can be defined in isolation. From a syntactic cartographer’s point of view, the ultimate goal of any investigation on predication is to prove if this link can be expressed in purely syntactic terms. This goal becomes particularly interesting when an axiom of the cartographic program is taken into account, namely that each linguistic feature, possibly associated with a morphosyntactic property, corresponds to a head in the functional hierarchy (Rizzi 2013: 435; Cinque and Rizzi 2015: 74):¹ is it possible to syntactically codify predication by means of a dedicated head – call it Pred° – corresponding to a ‘predicative feature’, namely a feature inherently associated with the subject-predicate link? This is the research question that will be addressed in the present chapter.

In the next section (§3.2) we shall see how a short survey of the main views on predication in the history of generative grammar can illuminate our research question. The remainder of the chapter includes: a section on how copular sentences represent an ideal case study to investigate predication (§3.3); a section on the interaction between

* This chapter is the result of a joint work entirely discussed and shared by both authors. However, Davide Mocci is specifically responsible for the reflections on the theory of predication in the nominal domain as developed in §§3.4.2-3.4.3. We are very grateful to two anonymous reviewers for their insightful and kind comments, and to Sam Wolfe for fuelling the values contained in this *Handbook*. We are also indebted to Ian Roberts, Guglielmo Cinque, and Luigi Rizzi for insightful comments on the proposal developed here.

¹ The matrix of this axiom can be identified in the principle of decompositionality, according to which universal grammar ‘imposes a maximum of one interpretable syntactic feature per lexical or functional element’ (Kayne 2005: 15). We are thankful to an anonymous reviewer for pointing this out to us.

predication and other structural domains, notably the nominal and inflectional domains (§3.4); finally, a section containing our concluding remarks (§3.5).

3.2 Historical notes

This section is devoted to surveying some of the main stands on predication in the history of generative grammar. We shall specifically concentrate on the change of perspective between the earlier works of Chomsky, where predication yielded an exocentric nuclear clause structure dubbed as S (the left periphery of the sentence being instead dubbed as S'), and Chomsky's *Barriers* (Chomsky 1986b), where instead predication was viewed as yielding an endocentric phrase (IP) like any other combination of words.

3.2.1 The combination of a property with an entity

The notions of sentence and predication, which are some of the many legacies handed down to us by ancient Greeks, are strictly related: sentences have been defined ever since Aristotle as those structures involving predication (Moro 2019: 1).² More explicitly, according to Aristotle a declarative sentence (*lógos apophantikós*) – which is the quintessential type of sentence – is a sequence of words which may be said to be true or false (Aristotle, *De interpretatione*, 4, 17a, 2–4). For instance, we can judge *they run* as true or false, but no such judgment is possible for *the run*; therefore, in keeping with Aristotle's definition, *they run*, as opposed to *the run*, is a sentence. Indeed, the crucial difference between *they run* and *the run*, which allows the former but not the latter to get a truth value, is – according to Aristotle – that *they run* involves the combination of a property with an entity; the property-denoting expression is dubbed as 'predicate' (*kategoroumenon*), whereas the entity-denoting expression is known as 'subject' (*hypokeímenon*).³ Thus, in *they run*, the property of running denoted by the predicate *run* combines with, i.e., is ascribed to, the (plural) entity denoted by the subject *they*. On the other hand, there is no combination of a property with an entity, and therefore no truth value whatsoever, in *the run*.⁴

Building on these insights by Aristotle, we may advance a working definition of predication: predication is the link between a predicate and a subject, where the predicate ascribes a property to the entity denoted by the subject. In this perspective, a sentence can be defined as the syntactic object consisting of the link between a subject and a predicate.

3.2.2 A formalization of predication

Generative grammarians both explicitly and implicitly adopted the Aristotelian view that a sentence consists of a link between a predicate and a subject (see Chomsky 1988: 54 for an explicit claim), and provided a precise formalization of such a view. Specifically, in a

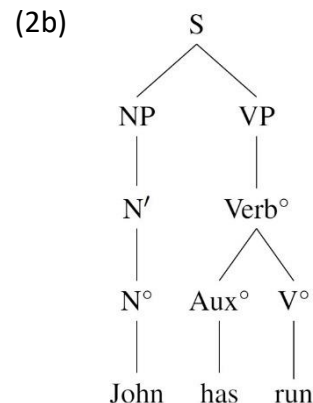
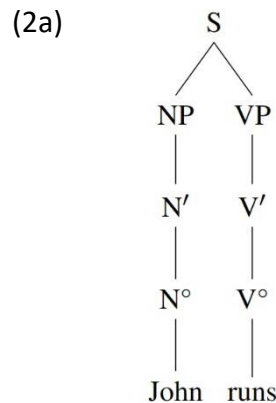
² Interestingly, both notions seem to be missing in the *Aṣṭādhyāyī*, the extremely advanced Sanskrit grammar attributed to the ancient Indian grammarian Pāṇini (ca. 4th century BC): see Mocci and Pontillo (2020) and the references cited therein.

³ 'For falsity and truth have to do with combination and separation' (Aristotle, *De interpretatione*, 1, 16a, 12–13), where what combines with or separates from something is a property, while the thing with which the property combines or from which it separates is an entity. See Moro (2017c: 19–20) for a comment on this passage. See Moro (2024: 25–29) on the reasons underlying Boethius's translation of *kategoroumenon* as 'predicate' (*praedicatum*), and Matthews (2019) for a general overview of Graeco-Roman grammars vis-à-vis modern linguistics.

⁴ Indeed, in the analytical philosophical traditions stemming from Frege and Russell's original reflections on language it is assumed that a predicate (i.e., property-denoting expression) combines with a variable inside noun phrases like *the run*. See Moro (2019) and §3.2.5 below for further details.

foundational work of generative grammar, Chomsky (1957[2002: 26-27, 39]) defined a sentence as in (1a), i.e., as the phrase generated by assembling a noun phrase (NP) with a verb phrase (VP), where VP may include an auxiliary (1b-c) (see also Chomsky 1956: 117-120).⁵ (1a-c) can be given a bidimensional representation, which corresponds to (2a-b) for *John runs* and *John has run*, respectively.⁶

- (1) a. $S \rightarrow NP VP$
 b. $VP \rightarrow Verb^{\circ} (NP)$
 c. $Verb \rightarrow (Aux^{\circ}) V^{\circ}$



In addition to the notion of sentence, the formalism instantiated in (2a-b) permits defining the notions of subject, predicate, and predication in purely syntactic terms (via configurations), thereby avoiding any reference to the semantic notions of entity and property: the subject is the NP immediately dominated by S; the predicate is the VP that is immediately dominated by S (Chomsky 1965[2015: 76-77]); predication, i.e., the subject-predicate link, is the sisterhood relationship holding between NP and VP (see Moro 2019). Another advantage of the formalism instantiated in (2a-b) is that it graphically captures a defining feature of sentences, namely their exocentricity. We shall discuss this point – which is of crucial importance for the understanding of predication – in the next subsection.

3.2.3 The exocentric nature of sentences

The issue of how to determine whether a phrase is endocentric (= headed) or exocentric (= headless) is riddled with controversies and cannot be addressed in any systematic or complete manner in this subsection. What we will be doing here is to apply the traditional distributivity test, which draws a sharp distinction between sentences and other phrases.

Thus, the distribution of an S like *John runs* (2a) does not coincide with the distribution of either NP (*John*) or VP (*runs*), as illustrated in (3):⁷

⁵ Chomsky (1981[1993: 25]) simplified the rewriting rules in (1a-c) as ' $S \rightarrow NP INFL VP$ ', where INFL corresponds to Aux° of (1) but is now represented as a sister to NP and VP.

⁶ We are abstracting away from the so-called 'affix hopping' rule in (2a), which is thus an over-simplification. This rule lets the third singular ending -s attach ('hop') to the verbal stem *run-*, thereby yielding *runs* (see Chomsky 1957[2002: 38-42, 62]).

⁷ Here we shall continue to refer to the phrase instantiated by *John* and the phrase instantiated by *runs* as NP and VP, respectively, in keeping with the terminology of Chomsky's classic texts (Chomsky 1956; 1957; 1965).

- (3) a. I love that [_S John runs].
 b. *I love that [_{NP} John].
 c. *I love that [_{VP} runs].

In this regard sentences contrast with NPs and VPs. For example, the distribution of the NP *John's runs* is the same as the distribution of the N° *runs*, inasmuch as the whole NP and *runs* can occur in the same syntactic environments:⁸

- (4) a. I love [_{NP} John's runs].
 b. I love [_{NP} runs].

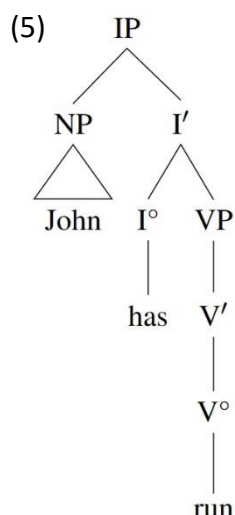
The element which, like the N° *runs* in (4), has the same distribution as the phrase of which it is a part is dubbed as the 'center' or 'head' of the phrase. Phrases that, like NP, are endowed with a head are endocentric; by contrast, phrases that, like S, lack a head are exocentric (see Bloomfield 1933: 194-196 and Zwicky 1985). In fact, the formalism deployed in (2a-b) makes perfect sense of the exocentricity of sentences (S): while immediately dominating NP and VP in (2a-b), S is a projection of neither NP nor VP, meaning that neither NP nor VP passes on its own category to S, which consequently does not qualify as either nominal or verbal. In current terms, both N° and V° fail to label S (see, e.g., Chomsky 2013: 42-43; 2019: 19).

In this way, the formalism developed by Chomsky's earlier work and exemplified in (2a-b) allows refining the definition of predication: predication is the combination of an NP (subject) with a VP (predicate) that yields an exocentric construction dubbed as "sentence". Put another way, predication is the combination of an NP with a VP neither of which projects to S (i.e., to the first node dominating NP and VP). In this perspective, it is not clear what the linguistic feature to be associated with predication is; equivalently, it is not clear what the syntactic head that – in keeping with the cartographic tenets – corresponds to such a feature is. We shall however see in the next subsections how this view of predication as the combination of an NP and a VP yielding an exocentric construction came to be replaced by a new approach to clause structure which bans exocentricity and introduces a linguistic feature to be associated with predication.

3.2.4 Going endocentric

In point of fact, sentences are the only phrases which are exocentric. To do away with this exceptional character of sentences, a new conception of clause structure was developed starting at least with Chomsky (1986b): sentences can be viewed as IPs, i.e., as inflection phrases headed by I° (inflection). This is illustrated in (5) for *John has run*.

⁸ See Adger (2013) for an updated discussion of the possible representations which could be assigned to complex NPs like *John's runs*. Let us note that the involvement of the same lexical items (i.e., *John* and *runs*) and the assignment of the same *theta*-role to *John* (i.e., agent) in (3) and (4) speak against an account of the different distribution of *John runs* vs. *John's runs* in terms of *theta*-role assignment (see Moro 2019: 2-3 for relevant discussion).



The core of this new conception is that, while inflection did not project any maximal category in the earlier generative models of clause structure (in fact inflection, represented as Aux° , was immediately dominated by Verb° in those models – see §3.2.2 above), it is promoted to the role of projector (I°) in the *Barriers* model (Chomsky 1986b), and thus projects its own maximal projection (IP) like any other heads. The conception of sentences as IPs came to be refined at the end of the Eighties, along the following lines.

By capitalizing on the peculiar agreement properties of inverse copular sentences (see §3.3 below), Moro (1988[2013: 19-41]) showed that I° should be split into Agr° (subject-verb agreement) and T° (tense). Independently Pollock (1989) arrived at the same conclusion by looking at the differential distribution of French finite vs. infinite verbs (see also Belletti 1990). This decomposition of I° into Agr° and T° , which goes under the rubric ‘Split-Infl hypothesis’, provided direct support for an axiom of syntactic cartography, namely that distinct linguistic features correspond to distinct heads in the functional spine of the clause (Cinque and Rizzi 2015: 67). Thus, a finite verb such as Italian *parl-av-o* ‘speak.IMPF.1SG’ can now be seen as the combination of three distinct heads: a V° head spelled out as *parl-* conveying the meaning ‘speak’; a T° head spelled out as *-av-* and bearing a tense feature (imperfect); finally, an Agr° head spelled out as *-o* and bearing a subject-verb agreement feature (first person singular).⁹

In this way, a sentence results as being a projection of $\text{Agr}^\circ + \text{T}^\circ$, with the subject sitting in the specifier of $\text{Agr}^\circ + \text{T}^\circ$ and the predicate being the complement to $\text{Agr}^\circ + \text{T}^\circ$ (6). In what follows we shall continue to refer to I° , but the latter should be taken as shorthand for the pair made up of Agr° and T° .

(6) [SUBJECT [I° (= $\text{Agr}^\circ + \text{T}^\circ$) [PREDICATE]]]

The representation in (6) shows that the inflectional features of the sentence (subject-verb agreement and tense) mediate the link between the subject and the predicate. This might suggest identifying the ‘predicative feature’ associated with the subject-predicate link with inflectional features, as has indeed been recently argued for by Cecchetto and Donati

⁹ The inflectional field of the clause was further expanded within the cartographic program in light of Cinque’s (1999) work on adverbs. See Rizzi (2013) for an insightful discussion.

(2023: 6).¹⁰ In this perspective, predication would be codified by (6), namely by a projection of I° (Chomsky 1986b). We shall however see in §3.3 below that this apparently intuitive codification of predication is empirically untenable: while inflection could be taken as a reflex of predication, predication may obtain even in the absence of inflection, so there must be other means in syntax to codify the subject-predicate link which do not involve I°.

3.2.5 Predication across categories: A dangerous misunderstanding

Before concluding this section, it may be useful to ponder on the way predication is used in formal semantics. Thus, formal semanticists assign an interpretation along the lines of (7) to the NP *the run*. The formula in (7) reads ‘there is a unique x such that x is a run’: this formula is a semantic structure, and is said to be formed at a derivational level – ‘logical form’ (LF) – which follows the syntactic level of the derivation, where structures such as *the run* and *they run* are built.¹¹

(7) $\lambda x. \text{run}(x)$

Crucially, the interpretation of *the run* given in (7) involves an instance of predication: the predicate *run* is predicated of a variable, namely x. This instance of predication we refer to as semantic predication, insofar as it pertains to LF only: the predicate *run* could not be predicated of variable x outside LF (i.e., outside (7)), because no variable shows up outside such a derivational level.

Semantic predication contrasts with syntactic predication. Thus, (8a) features an instance of semantic predication at LF: the predicate *run* is predicated of x in keeping with (7). On the other hand, (8b) features an instance of syntactic predication: the predicate *run* is predicated of *they* independent of the interpretation that is assigned to *they run* at LF. Crucially, only a syntactic instance of predication may occur as the complement of *that* in (8): whence the well-formedness of (8b) as opposed to (8a). If there were no difference between semantic and syntactic predication, the contrast in (8) would be hard to explain (see Moro 2017c: 249 n. 28; 2019: 2).

- (8) a. *John says that the run.
b. John says that they run.

Thus, semantic predication must be kept distinct from syntactic predication. The kind of predication we are concerned with in this chapter is syntactic predication.

3.2.6 Summary

Predication – more precisely, syntactic predication – is the link between a subject and a predicate in a clause structure. In the present section we considered two views of predication, corresponding to two conceptions of clause structure. According to the first view, going back at least to Chomsky (1956), (1957), there is no element which mediates between the subject (NP) and the predicate (VP), inflection (Aux°) being a part of the predicate: the subject-predicate link is codified in syntax by assembling an NP and a VP into

¹⁰ To be specific, Cecchetto and Donati inherently associate predication with subject-verb agreement, which they refer to as ‘feature-sharing’.

¹¹ See Partee, ter Meulen, and Wall (1990); Heim and Kratzer (1998); Chierchia and McConnell-Ginet (2000) for a detailed discussion.

an exocentric structure (dubbed as ‘S’), i.e., a structure that is not projected by any element. This view of predication as yielding S we dub as the ‘exocentric view of predication’, in that it corresponds to a conception of sentences as exocentric phrases.

On the other hand, according to the second view, which may be traced back at least to Chomsky (1986b), predication is codified by IP, whose head (I^0) mediates the link between the subject and the predicate. We dub this second view as the ‘endocentric view of predication’, in that it corresponds to a conception of sentences as endocentric phrases.

3.3 An ideal case-study: Copular sentences

Copular sentences constitute an ideal case-study to investigate predication, inasmuch as they factor out a fundamental morphological asymmetry which characterizes sentences with a lexical predicate: whereas the subject and the predicate are realized as an NP and a VP, respectively, in sentences with a lexical predicate, the subject and the predicate can both be realized as NPs in copular sentences. Thus, in this section we shall see how copular sentences force us to abandon the traditional views of predication (surveyed in §3.2 above), while at the same time providing a fresh empirical perspective on the syntactic codification of predication.

3.3.1 A dogma in the theory of clause structure

In the previous section (§3.2), we considered two views of predication, which we referred to as the exocentric and the endocentric view of predication. These views differ in that the structure resulting from assembling the subject with the predicate is an IP in the endocentric view (9), but an exocentric phrase (dubbed as ‘S’) in the exocentric view (10). Let us however emphasize that the two views of predication crucially share the following property: that the subject can be defined configurationally as the NP that is immediately dominated by the sentential node (i.e., IP or S) in both (9)-(10).

(9) [IP NP_{SUBJ} [I' I^0 [VP_{PRED}]]]

(10) [S [NP_{SUBJ}] [VP_{PRED}]]

This definition originally had the status of a postulate. However, Chomsky (1986a: 116), building on Rothstein (1983), attempted to reduce this definition to a theorem by deriving it from an independent logical principle, namely function saturation. Chomsky’s attempt may be broken down into two parts for expository purposes. Firstly, he adopted the Fregean and Montagovian view that a predicate – e.g., *runs* such as in *John runs* – is the linguistic correlate of an unsaturated logico-mathematical function, represented as *run(x)*, and that a subject – e.g., *John* such as in *John runs* – is instead the linguistic correlate of the argument which saturates (i.e., assigns a value to the variable bound by) that function; in this perspective, the combination of *John* and *runs* into a sentence (*John runs*) is represented as a saturated function, namely ‘*run(John)*’.¹² Secondly, making a step forward with respect to this Fregean and Montagovian view, Chomsky identified the *locus* of function saturation with the NP immediately dominated by S/IP. In this way, the so-called ‘extended projection principle’ (EPP), namely the requirement that an NP be inserted into a

¹² An explicit acknowledgement of Chomsky’s intellectual debt to Frege can be found in Chomsky (1986a: 116). See Montague (1973) and Dowty, Wall, and Peters (1981), along with the historical remarks in Moro (2017c: 56-57), for the analysis of predication developed within Montague’s grammar.

position immediately dominated by S/IP, follows from a general logical requirement: that a function (denoted by the predicate, i.e., VP) be saturated by an argument of the appropriate type (namely an NP).¹³

The configurational definition of subject as the NP immediately dominated by S/IP, now viewed somewhat as a theorem, became a sort of dogma of generative grammar (Lepschy 2000: 84). Indeed, copular sentences proved such a dogma wrong both empirically and theoretically, with dramatic consequences for the theory of predication, as we shall see in the next subsections.

3.3.2 The anomaly of inverse copular sentences

Copular sentences are sentences whose main verb is a copula, namely a token of *be*. Here we shall focus on copular sentences of the ‘NP *be* NP’ type, exemplified in (11).

- (11) a. [_{NP} A picture of the wall] was [_{NP} the cause of the riot].
 b. [_{NP} The cause of the riot] was [_{NP} a picture of the wall].

Asymmetries like those in (12) (from Italian) and (13) reveal that (11a-b), despite being superficially very similar, indeed have very different structures (Longobardi 1985; Moro 1997a; 2017a). Thus, the agreement asymmetry in (12) can be captured by taking the pre-copular NP to be a subject only in (12a), coupled with the principle that the verb – including the copula – always agrees with the subject in Italian.

- (12) a. Queste foto del muro sono
 this.PL picture.PL of.the wall be.3PL
 la causa della rivolta.
 the.SG cause.SG of.the riot
 ‘These pictures of the wall are the cause of the riot’
- b. La causa della rivolta sono/*è
 the.SG cause.SG of.the riot be.3PL/be.3SG
 queste foto del muro.
 this.PL picture.PL of.the wall
 ‘The cause of the riot is (lit. “are”) these pictures of the wall’.

The asymmetry in (13) instead follows from an independently motivated locality principle: the subjacency condition à la Cinque (1990). Referring the reader to Moro (1997a: 49-57; 2017a) for a detailed discussion, here we may limit ourselves to noting that arguments of a predicate are subject to the subjacency condition, whereas predicates are not (Moro 2017c: 163). Since the post-copular NP is a predicate in (13a), but a subject (i.e.,

¹³ In this perspective, *there* of (ib) was interpreted as an expletive of the subject, i.e., as an element that is inserted in the subject position only when that position is not already being filled by the genuine subject *a man* (see Chomsky 1986a: 115-116; cf. Moro 2019: 3 for critical considerations).

- (i) a. A man is in the room.
 b. There is a man in the room.

For an alternative interpretation of the data in (i), whereby *there* is an expletive of the predicate (*in the room*) rather than of the subject, see Moro (1997a: 94-166); (2017b); (2017c: 171-185).

an argument of the predicate) in (13b), the extraction of the *wh*-phrase from the post-copular NP is sensitive to – and blocked by – the subadjacency condition only in (13b).¹⁴

- (13) a. [Which riot]_i do you think a picture of the wall was the cause of *t_i* ?
 b. *[Which wall]_i do you think the cause of the riot was a picture of *t_i* ?

All in all, the asymmetries in (12)-(13) reveal that (11a) has a different structure than (11b). Specifically, (11a) is a canonical copular sentence, i.e., a sentence in which the pre-copular NP fulfills the function of subject, while (11b) is an inverse copular sentence, i.e., a sentence in which the pre-copular NP fulfills the function of predicate (see Moro 1988; 1997a; 2017a; 2017c).

Crucially, Moro proposed that canonical and inverse copular sentences are the two different manifestations of a single basic structure, depicted in (14a); here NP_{SUBJ} and NP_{PRED} form a phrase (α) whose exact label we suspend for the time being. When NP_{SUBJ} moves to Spec-IP (14b), canonical copular sentences such as (11a) are derived; by contrast, when NP_{PRED} moves to Spec-IP (14c), inverse copular sentences such as (11b) are derived.¹⁵

- (14) a. [IP__ I° [VP *be* [α NP_{SUBJ} NP_{PRED}]]]
 b. [IP NP_{SUBJ} I° [VP *be* [α *t* NP_{PRED}]]] (canonical)
 c. [IP NP_{PRED} I° [VP *be* [α NP_{SUBJ} *t*]]] (inverse)

The discovery of inverse copular sentences has a dramatic impact on the traditional theory of predication. To understand this point, we have to reconsider what we referred to as a dogma in the theory of clause structure, namely the requirement that Spec-IP be filled by the subject (see §3.3.1 above).

In copular sentences like (11), Spec-IP may be occupied by either one of the subject and the predicate. It thus turns out that the allegedly logical requirement that Spec-IP be filled by the subject is an illusion: the only constraint imposed on Spec-IP is that this position be occupied by a category belonging to the morphological class of nouns, i.e., NP;¹⁶ since both the subject and the predicate are categories of the NP type in (11), both can fill Spec-IP. All in all, copular sentences constitute an ideal case-study to investigate the syntactic codification of predication because they allow us to factor out aspects that are completely orthogonal to the subject-predicate link, such as the categorial distinction between NPs and VPs (Moro 1997a; 2019).

¹⁴ See Moro (1997a), (2017a) for other asymmetries between (11a) and (11b).

¹⁵ The class of inverse copular sentences also includes *there*-sentences. See Moro (1997a); (2017b); (2017c); Hale and Keyser (2002: 189-203) for a detailed discussion.

¹⁶ Cf. the ill-formedness of (i)-(ii), where the embedded Spec-IP is occupied by an adjectival phrase (*tall*) and a VP (*runs*), respectively. The requirement for Spec-IP to be occupied by an NP is only apparently contradicted by the well-formedness of (iii), an example of ‘locative inversion’ borrowed from Hoekstra and Mulder (1990: 31).

- (i) *I think that tall is John.
 (ii) *I think that runs John
 (iii) Into the room walked a man.

Evidence from several domains of syntax (Moro 1997a: 289-90 n. 12; den Dikken 2006: 98-101) suggests that the prepositional phrase *into the room* sits not in Spec-IP, but rather in the left periphery of the clause, specifically in the position where topics are hosted (den Dikken 2006: 102). We are thankful to an anonymous reviewer for drawing our attention to this point.

3.3.3 Where inflection fails to mediate the subject-predicate link

Crucially, when Spec-IP is occupied by the predicate, as in inverse copular sentences (14c), I° can hardly be taken to mediate the link between the subject and the predicate: the link which I° is mediating in (14c) is that between a predicative NP (sitting in Spec-IP) and a VP (sitting in the complement of V°), but this VP cannot be considered as a predicate, because the grammatical function of predicate is already fulfilled by the NP sitting in Spec-IP. Of course, VP cannot possibly be considered as a subject either, because the grammatical function of subject is already fulfilled by the α -internal NP. All in all, I° fails to mediate the link between a subject and a predicate in inverse copular sentences, because neither one of the slots projected by I° (namely Spec-IP and the complement to I°) contains the subject. This disqualifies IP from being the logical unambiguous scaffolding of the subject-predicate link; in other words, the syntactic codification of predication by means of IP is too restrictive, in that it cannot cover inverse copular sentences. We conclude from this that inverse copular sentences constitute compelling evidence against any theory of predication that: i) identifies the subject-predicate link with some inflectional feature (notably, tense and agreement); and ii) codifies the subject-predicate link by means of a projection of I°.

Additional empirical evidence against an IP-based codification of predication comes from languages like Russian which exploit an agreement mark on adjectives to distinguish between a predicative and a non-predicative structure (Jespersen 1924: 121, also quoted in Moro 1997a: 54; see Moro 2000: 36 for similar examples from German):

- (15) a. dom novyi
 house new.AGR
 ‘the new house’ (NON-PREDICATIVE STRUCTURE).
 b. dom nov
 house new
 ‘the house is new’ (PREDICATIVE STRUCTURE).

Crucially, the predicative structure is the one in which *nov* ‘new’ (the adjectival predicate) is not marked for agreement with *dom* ‘house’ (the subject). This shows that the subject-predicate link cannot by any means be associated with agreement – nor with any other inflectional features – in languages like Russian. Consequently, the *prima facie* innocuous hypothesis that the subject-predicate link is codified in syntax by IP is untenable on comparative grounds, inasmuch as it cannot extend to languages like Russian (Moro 1997a: 54).¹⁷

¹⁷ Relevant in this connection are also “rich small clauses” (RSC): predication obtains between *gli scacchi* (subject), which is masculine plural, and *la mia passione* (predicate), which is instead feminine singular, in RSC (i), with no agreement whatsoever between the subject and the predicate (Moro 1997a: 53; 2017c: 248 n. 21).

- (i) Ritengo [RSC [gli scacchi] [la mia passione]].
 regard.PRS.1SG the.PL.M chess.PL.M the.SG.F my.SG.F passion.SG.F
 ‘I consider chess my passion’.

RSC can be exploited to show that predication is orthogonal to case agreement, too. Thus, the RSC-internal subject (*Yuriya Solomina*) is accusative-marked whereas the RSC-internal predicate (*talantlivym aktërom*) is instrumental-marked in the following Russian example from Pereltsvaig (2007: 100) (see Comrie 1997 for a crosslinguistic perspective on the case marking of predicates).

In sum, inverse copular sentences and Russian sentences with an adjectival predicate converge in showing that predication may obtain even in the absence of inflection (i.e., of tense and agreement), and thereby constitute evidence against the identification of the predicative feature with an inflectional feature, as well as against the codification of predication by means of a projection of I°. ¹⁸ Indeed, it should be noted that inverse copular sentences also constitute evidence against the earliest generative model of clause structure where predication was codified as 'S', i.e., as the category formed by combining an NP with a VP (see §3.2 above). Thus, in keeping with that model, the amalgam formed by the predicative NP (*the cause of the riot*) and VP (*was a picture of the wall*) should instantiate predication in (16), but this is not the case: VP is not a predicate here, so the combination of NP with VP cannot constitute a subject-predicate link by definition.

(16) [_S [_{NP} The cause of the riot] [_{VP} was a picture of the wall]]

However, this is not the end of the quest for the predicative feature, namely the linguistic feature to be inherently associated with the subject-predicate link. We shall see how this quest was pursued in novel ways by Bowers (1993) – and others who built on his insights – in the following subsection.

3.3.4 The quest for a predicative feature

Given that the subject-predicate link is not codified at the IP level (§3.3.3), the question arises as to where exactly the subject-predicate link is established. When it comes to copular sentences – whose basic structure is repeated below as (17) – the null hypothesis is that the subject-predicate link is established at (the as yet unspecified level) α , i.e., at the first level where the NP subject co-occurs with the predicative NP. Let us therefore assume that predication is codified in syntax by α .

(17) [_{IP}__ I° [_{VP} *be* [_{α} NP_{SUBJ} NP_{PRED}]]]

At this point one may wonder whether α contains a head which mediates the link between NP_{SUBJ} and NP_{PRED} in (17). In fact, in keeping with the cartographic guidelines, a head could be posited that links NP_{SUBJ} with NP_{PRED} as in (18), provided it corresponds to a linguistic feature. Thus, cartographic work on predication such as Rizzi (2015) and Shlonsky and Rizzi (2018) assumes a Pred° head bearing a predicative feature; the semantic contribution of such a feature may be viewed as an instruction to the effect that a subject (namely, Spec-PredP) should be linked with its predicate (namely, the complement of Pred°). ¹⁹

-
- (ii) Ja ne sčitaju [_{RSC} [Yurija Solomina] [talantlivym aktërom]].
 I not consider Yury.ACC Solomin talented.INS actor.INS
 'I do not consider Yury Solomin a talented actor'.

¹⁸ Thus, Cecchetto and Donati's (2023: 6) view that predication is inherently associated with agreement (feature-sharing) appears to be empirically untenable.

¹⁹ See Bowers (1993: 595). Den Dikken (2006) similarly assumes a head – dubbed as 'Relator' – that negotiates the link between a subject and a predicate, but does not identify such a head with any specific functional category: Relator is a placeholder for any functional head, which means that Relator may be instantiated by I°, by P°, by Focus°, by Topic°, etc., but does not indeed coincide with any of these heads. See especially den Dikken (2006: 15-16) in this connection.

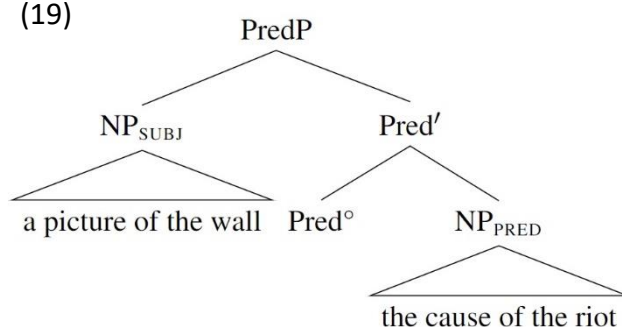
(18) [IP__ I° [VP *be* [PredP NP_{SUBJ} [Pred° NP_{PRED}]]]]

All in all, the structure in (18) is perfectly in the spirit of syntactic cartography: this structure complies with the cartographic axiom that one linguistic feature corresponds to one functional head, and comports with the tendency – which we have seen as dominant in generative grammar since at least Chomsky (1986b) (see §3.2 above) – to treat all phrases as endocentric. Nonetheless, it may be worth asking whether the postulation of a predicative feature can be supported on independent grounds.

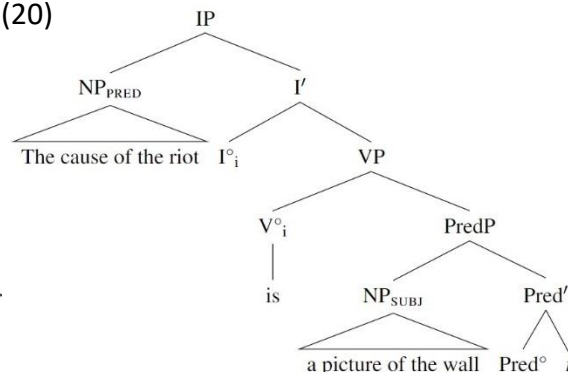
3.3.5 Some consequences of PredP

To start with, predication is associated with a specific syntactic configuration (19) in a model that adopts PredP. Now, the configuration in (19), being a specifier-head-complement configuration, is non-distinct from the configuration formed by assembling I° with *the cause of the riot* (Spec-IP) and the VP *is a picture of the wall* (complement of I°) in (20), which is the phrase-marker of an inverse copular sentence. Crucially, however, the configuration formed by assembling I° with Spec-IP and the complement of I° in (20) does not by any means codify a subject-predicate link: as established in §3.3.3, the latter configuration codifies a relationship between a predicate (the NP in Spec-IP) and a VP which qualifies neither as a predicate (because the grammatical function of predicate is already fulfilled by the NP in Spec-IP) nor as a subject (because the grammatical function of subject is already fulfilled by the PredP-internal NP). Therefore, following Moro (2019: 3), we conclude that in a model that adopts PredP the configuration that codifies predication (19) cannot be differentiated from a configuration that does not codify predication (i.e., IP of (20)).

(19)



(20)



Indeed, what really distinguishes a predicative structure (19) from a non-predicative structure (the IP of (20)) is not a syntactic configuration in a PredP-based model of predication; rather, the crucial factor distinguishing between (19) and (20) is the fact that a specific head (Pred°) – or, equivalently, a specific feature ('predicative feature') – is more appropriate than others (I°) to connect a subject with its predicate. Crucially, the distinction between the heads (or features) which are appropriate to establish the subject-predicate link and those which are not likewise appropriate cannot be traced back to any independently motivated syntactic principle: it is merely a lexical fact (more precisely, a fact pertaining to the functional lexicon) that Pred° (or, equivalently, the predicative feature corresponding to Pred°) allows a subject to be linked with a predicate, while, e.g., I° (or, equivalently, the inflectional features corresponding to I°) does not. Consequently, the

question as to how predication is codified in syntax reduces to a mere lexical matter in a model that adopts PredP, thereby undermining the goal of unambiguously associating predication to a dedicated syntactic configuration.

Furthermore, we would like to underscore the fact that PredP (19) is an endocentric phrase. Therefore, we predict that Pred°, being the head of PredP, has the same distribution as PredP. However, this prediction is false, for principled reasons which we shall now examine. Indeed, predication (realized as a PredP) is a link between two words (more precisely, two phrases, namely the subject and its predicate). The reflex that the subject-predicate link has been established in a sequence of words is the fact that we can judge that sequence of words as true or false (see §3.2.1 above). A single word can never be judged as true or false (modulo the generation of silent pronouns),²⁰ and never instantiates the subject-predicate link. Accordingly Pred°, which is a single word, cannot in itself instantiate predication (PredP) by definition, thereby falsifying the prediction that PredP has the same distribution as Pred°.

Of course, one may appeal to the selectional requirements of Pred° to account for the impossibility of a single word to codify predication: Pred°'s selectional requirements are frustrated whenever Pred° does not co-occur with a subject (which is allocated to Spec-IP) and a predicate (which is allocated to the complement of Pred°). However, this appeal to the selectional requirements of Pred°, far from being a genuine explanation (i.e., the reduction of a complex empirical phenomenon to an independently motivated principle), is merely the reproduction, with other words, of what we have already observed (namely, that predication is the link between a subject and a predicate).

In sum, the postulation of a predicative feature corresponding to Pred° may *prima facie* appear to be consonant with the cartographic program, an axiom of which is that each linguistic feature corresponds to a syntactic head. However, more careful examination reveals a very different picture. First, the postulation of a predicative feature and of its corresponding head reduces the question as to how predication is codified in syntax to a mere lexical matter (Pred° has an appropriate featural makeup which other heads lack), thereby undermining the goal of codifying predication in purely syntactic terms. Second, the postulation of a PredP, by implying that a single word (Pred°) has the same distribution as the whole subject-predicate link (PredP), fails to capture the fact that no single word can instantiate predication.

Therefore, predicative features and the Pred° head do not really constitute a genuine expression of the cartographic research program, insofar as they hinder a major goal of this program, namely the unambiguous association of linguistic (morphosyntactic, syntactic, semantic, pragmatic) phenomena to dedicated syntactic configurations (see, among many others, Rizzi 1997; Cruschina and Bianchi 2021: §2.2; Shlonsky and Bocci 2019). To reach this goal and thus underpin the cartographic research program, an alternative codification of predication in syntax is called for. We shall discuss such an alternative codification in the next subsection.

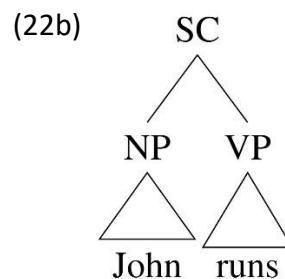
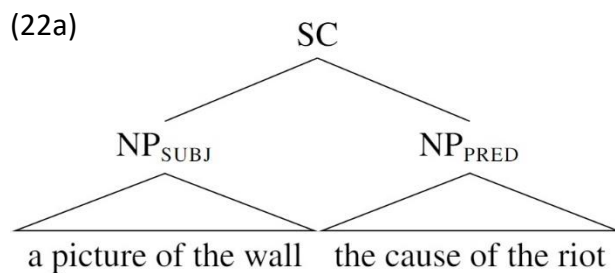
3.3.6 Symmetry in syntax

The exocentric nature of predication – i.e., the fact that no single word can instantiate predication – could be naturally captured, in X-bar-theoretic terms, by taking the phrase

²⁰ Thus, Italian *arrivano* 'arrive.3PL', which can be judged as true or false, is a two-word combination involving a silent pronoun (*pro*) at the relevant level of representation: *pro arrivano*. See Rizzi (1982) and Moro (2019: 6 n. 7).

codifying the subject-predicate link to be headless. We already know from the domain of copular sentences (see §3.3.3 above) that such a phrase should be identified with a structural level α lower than the one where inflection is codified (i.e., lower than the level referred to as S in the earlier stages of generative grammar, and as IP, AgrP, or TP in the subsequent stages). Let us therefore assume that α is an exocentric phrase in (21). To be specific, α is a ‘bare small clause’ (SC), i.e., a phrase that merely consists of a subject and a predicate, which are not linked by any elements (e.g., T° and Agr°): see (22a-b).²¹

- (21) a. [_{IP} __ I° [_{VP} *be* [_{α} NP_{SUBJ} NP_{PRED}]]]
 b. [_{IP} __ I° [_{α} NP VP]]



Thus, SC is a symmetrical structure, in the technical sense that the components of SC (namely, the subject and the predicate) mutually c-command one another and neither projects (Moro 2000; Moro and Roberts 2024). At this point, we adopt Moro’s (2019: 4) conjecture, reported below (see also Moro 2000: 71). This conjecture makes it possible to handle the notion of predication in purely geometrical terms, i.e., by making reference to the configurations created by assembling lexical items on a bidimensional space: the subject-predicate link is now reduced to a configuration, i.e., (22), with no reference to semantic notions, such as entity and property, which are hard to pin down.²²

- (23) Moro’s conjecture: All and only predicative structures are codified in syntax by symmetrical constructions of the XP YP type where neither XP nor YP projects.

On a historical note, symmetry had already been deployed to codify predication in the earlier generative models, where the subject-predicate link was represented as ‘S’ (see §3.2 above). Symmetry was subsequently banned from syntax in later models, where constructions which had previously been analyzed as symmetrical phrases – such as sentences and double-object constructions – were reduced to asymmetrical phrases (see the works of Chomsky 1986b and Larson 1988). However Moro, noting that symmetrical phrases are perfectly compatible with the standard definition of Merge provided by Chomsky (1995: 243-248), argued in the late Nineties that Merge should indeed be

²¹ Bare small clauses (SC) differ from rich small clauses (RSC) in that RSCs involve a silent head intervening between the subject and the predicate. See Moro (2000: 40-48), where a series of empirical diagnostics are provided to detect the presence of this silent head.

²² The conjecture in (23) is reminiscent of Williams’s (1980) constraint that all and only subjects mutually m-command their predicates (see also Hale and Keyser 2002: 249 n. 2). Note however that Williams’s constraint would have the undesirable consequence of forcing the phrase in Spec-IP to be predicated of the phrase in the complement of I°. On the empirical motivations that make such a consequence undesirable, see the discussion of inverse copular sentences in §3.3.3 above.

permitted to form symmetrical phrases (Moro 1997b; 2000: 31-38). Recently Chomsky (2013: 43) and Rizzi (2015: 21) adopted Moro's symmetrical version of Merge, thereby reintroducing symmetry into the standard apparatus of generative grammar.

Our quest for the predicative feature terminates here. There is no predicative feature and no Pred° head corresponding to such a feature. All that is needed to codify predication in syntax is the general compositional algorithm Merge: Merge creates an exocentric phrase dubbed as SC, which can be defined by resorting to the purely geometrical notion of symmetry; a subject-predicate link is nothing more than SC. By unambiguously associating predication to SC, this approach contributes to the cartographic enterprise, a major goal of which is the unambiguous association of linguistic phenomena to dedicated syntactic configurations.

3.3.7 Summary

Data from inverse copular sentences and Russian sentences involving a predicative adjective show that predication is established at a level lower than IP. We have rejected the hypothesis that this lower level is a PredP on empirical and conceptual grounds. On the other hand, following Moro (2000; 2019), we have identified the lower level at stake with a 'bare small clause' (SC), namely with an exocentric structure defined by means of the configurational notion of symmetry, thereby reintroducing exocentricity into the clausal spine. In this way, predication can be handled in purely geometrical (*qua* configurational) terms, which ultimately supports the cartographic enterprise: predication obtains only when SC is formed.

In the next section, we would like to ponder on a consequence of the analysis outlined above: the role of IP – as well as of other functional projections – in the model.

3.4 Syntactic epenthesis

We saw in the previous section that IP cannot be taken as codifying predication in syntax. What is the function of IP then? The SC-based model of predication offers a fresh perspective on this question, while also shedding new light on functional projections in the nominal domain. Specifically, we shall argue that IP is the syntactic equivalent of epenthesis in phonology.

3.4.1 Syntactic epenthesis in the clausal domain

Any syntactic structure must satisfy the two requirements imposed by the linguistic interfaces assumed in the minimalist model: the linearization requirement – which is imposed by the sensory-motor interface – as well as the labeling requirement, imposed by the conceptual-intentional interface. Now, the algorithms that provide for linearization and labeling are formulated in such a way as to be satisfied in asymmetrical structures only.²³ Crucially, the SC depicted in (22) (see above) is a symmetrical structure, and therefore cannot be labeled or linearized. This means that SC fails to satisfy the requirements imposed by the two linguistic interfaces, and hence constitutes an unstable syntactic structure.

In the theory of generalized dynamic antisymmetry (Moro and Roberts 2024), illegitimate symmetrical structures such as SC can indeed be generated, provided that movement intervenes to restore asymmetry into them before they reach the surface level of

²³ The linearization algorithm, dubbed as 'Linear Correspondence Axiom', was originally formulated by Kayne (1994: 3-6). The standard formulation of the labeling algorithm is instead due to Chomsky (2013: 43-44) (for a refined formulation based on intervention locality, see Rizzi 2016).

language (so-called PF, i.e., phonetic form). For instance, the movement of either component of SC to Spec-IP turns SC into an asymmetrical structure, thereby allowing for both the linearization and labeling of SC. This is illustrated by canonical and inverse copular sentences (see §3.3.2 above):

- (24) a. [IP ___ I° [VP *be* [SC NP_{SUBJ} NP_{PRED}]]]
 b. [IP NP_{SUBJ} I° [VP *be* [SC *t* NP_{PRED}]]]
 c. [IP NP_{PRED} I° [VP *be* [SC NP_{SUBJ} *t*]]]

In this perspective, the insertion of IP on top of SC is like the insertion (‘epenthesis’) of into the Ancient Greek string **a-mrotos* (lit. non-mortal, hence ‘immortal’): this string is unstable because it contains an illegitimate consonantal cluster, namely <mr>; the insertion of in between <m> and <r> rescues this string by interrupting the illegitimate cluster <mr> and thus yields the well-formed string *ámbrotos*. Likewise, IP rescues SC by virtue of containing an empty specifier (Spec-IP), which permits either one of NP_{SUBJ} and NP_{PRED} to move outside SC, thereby interrupting the illegitimate symmetrical structure formed by these NPs (Moro 2019: 4-5).

Sticking to a proposal by Moro (2023), where the insertion of functional material to solve the symmetry between two maximal projections was dubbed as ‘syntactic epenthesis’, we take IP to be an instance of syntactic epenthesis. Among the advantages of this view is that it allows us to derive the universal intermediary position of IP as a theorem: since the Comp-field (i.e., the structural domain devoted to codifying scope-discourse properties) is universally higher than SC (i.e., the structural domain devoted to codifying predication), and since IP is always inserted on top of SC to interrupt the illegitimate symmetry of SC, it follows that IP must be universally sandwiched between the Comp-field and SC (see Moro 2019: 5).²⁴

We shall see in the next subsection that other instances of syntactic epenthesis exist in syntax, notably in the nominal domain.

3.4.2 Syntactic epenthesis in the nominal domain

It has long been assumed that predication obtains even within NPs. For instance, *cretina* ‘idiot’ would be a predicate and *Gianna* a subject in the Italian complex NP (25) (see Napoli 1989). This assumption was made formally explicit by Kayne (1994: 106) and den Dikken (2006: 161-246). However, here we shall abstract away from the technical details of their proposals, and shall instead focus on the consequences that the SC-based codification of predication has for the analysis of complex NPs like (25).

- (25) Quella cretina di Gianna
 that.SG.F idiot.SG.F of Gianna
 ‘That idiot of a woman whose name is Gianna’.

²⁴ This leaves open the question as to whether the number and typology of phrases populating the Comp and inflectional fields are uniform across languages or vary depending on the language-specific repertoire of features. Evidence in favor of the latter possibility is provided by Moro’s (2011) study on multiple coordinated *wh*-phrases.

To start with, *di* may be taken as playing the same role played by I° in the clausal domain: *di* projects an empty specifier to which one of two components of SC (i.e., *cretina*) moves, as illustrated below.²⁵

- (26) a. quella [DP ___ [D° di [SC [NP Gianna] [NP *cretina*]]]]
 b. quella [DP [NP *cretina*] [D° di [SC [NP Gianna] *t*]]]

Thus, the phrase projected by *di* (DP) constitutes the minimal structural space needed to interrupt SC – i.e., the illegitimate symmetrical structure which codifies the link between the subject (*Gianna*) and the predicate (*cretina*) – via movement. Put another way, DP is an instance of syntactic epenthesis just like IP.²⁶

However, the DP of (26) differs from IP in one important respect: the DP of (26) displays an endocentric distribution, in the sense that the distribution of DP coincides with the distribution of determiners such as *me* ‘me’ (27);²⁷ by contrast, IP has an exocentric distribution, in the sense that the distribution of IP does not coincide with the distribution of any of the words dominated by IP (see §3.2 above).

- (27) Quella [DP *me*] non esiste più.
 that.SG.F me not exist.PRS.3SG anymore
 ‘That version of me does not exist anymore’.

We shall not attempt a fully developed account of this contrast between DP and IP, which remains beyond our understanding at present. Our more limited goal here is to suggest what a possible line of analysis could look like, leaving the development of this line of analysis for future work.

I° has been argued to be defective (Chomsky 1986b). If we take the defectivity of I° to mean that I° lacks its own distribution, then I° would be forced to borrow the distribution of the first non-defective category dominated by I°. Such a category is not the VP projected by the copula (*be*) in either (28a) (i.e., the schematic representation of sentences in which the

²⁵ DP (‘determiner phrase’) is the phrase headed by a determiner (D°) such as an article, a pronoun, etc.

²⁶ This analysis of DP has been verified in several languages and constructions, including Sanskrit compounds (Mocci 2022; 2024). As pointed out to us by an anonymous reviewer, another good candidate in which this analysis could be fruitfully tested is Chinese Adjective-Noun sequences involving *de* (Cinque 2010: 95-97).

²⁷ The ill-formedness of (i) may be interpreted as evidence that the D° *di* does not indeed have the same distribution as the whole DP *cretina di Gianna*.

- (i) *Quel/quella [DP *di*].
 that.SG.M/SG.F of

While this interpretation cannot be readily accommodated within our account, an alternative possibility may be advanced whereby the ill-formedness of (i) does not stem from distributional factors: *di* is unwarranted in (i) just like *to be* in (iic), because there is no small clause for which *di* and *to be* can serve as an epenthesis (see Moro 1997: 37 and den Dikken 2006: 143-148 for a detailed discussion of (ii)).

- (ii) a. John considers a picture of the wall (to be) the cause of the riot].
 b. John considers the cause of the riot *(to be) a picture of the wall].
 c. *John considers to be.

We are thankful to an anonymous reviewer for suggesting this alternative possibility.

predicate is realized by a lexical verb) or (28b) (i.e., the schematic representation of copular sentences). This is because, even when the copula sits under V° such as in (28b), it is nothing more than the spell-out of inflection (specifically, of tense) along the lines of Aristotle, and hence qualifies as defective just like I°.²⁸ Therefore, the first non-defective category dominated by I° is SC not only in (28a), but in (28b) as well.

- (28) a. [IP__ I° [SC NP VP]]
 b. [IP__ I° [VP *be* [SC NP_{SUBJ} NP_{PRED}]]]

Since SC, which is an exocentric phrase, has an exocentric distribution, IP – which inherits its distribution from SC – will also end up displaying an exocentric distribution in both (28a) and (28b). On the other hand *di*, not being defective, fails to inherit the exocentric distribution of SC in (26); consequently, the phrase projected by *di* (DP) displays an endocentric distribution.

3.4.3 Summary

The codification of predication via bare small clauses (SC) opens up a novel perspective on the inflectional and nominal domains: neither I° nor elements like Italian *di* mediate the link between the subject and the predicate, which is established at SC; rather, I° and *di* project a phrase (IP and DP, respectively) which constitutes the minimal structural space needed to interrupt an illegitimate symmetrical structure (SC). Accordingly, IP and DP are instances of syntactic epenthesis.

3.5 Conclusion

In this chapter we searched for a purely syntactic codification of predication (i.e., the link between a subject and a predicate), in keeping with the cartographic goal of unambiguously associating linguistic phenomena to dedicated syntactic configurations. We considered two possibilities to this effect.

A first possibility, suggested by the cartographic axiom ‘one feature – one head’ (Cinque and Rizzi 2015: 74), is that predication is codified in syntax by a PredP, i.e., an endocentric phrase whose head (Pred°) bears a predicative feature. However, it was proved that this solution is untenable for empirical and theoretical reasons: i) the postulation of a predicative feature and of its corresponding head (Pred°) reduces the question as to how predication is codified in syntax to a mere lexical matter, thereby undermining our objective of codifying predication in purely syntactic terms; ii) the postulation of a PredP makes the prediction that predication can be instantiated by a single word (Pred°), but this prediction is never borne out, for principled reasons: predication is by definition a compositional notion, one which obtains when two distinct phrases (the subject and the predicate) are combined.

We subsequently tested a second possibility, namely that predication is codified by an exocentric phrase, dubbed as ‘bare small clause’ (SC): SC is a phrase in which the subject and the predicate mutually c-command one another and neither projects. This second possibility, which reintroduces exocentricity into the clausal spine, makes it possible to handle predication in purely geometrical (*qua* configurational) terms, with no recourse to

²⁸ For a historical overview of the copula, which goes from Aristotle to Jespersen passing through Abelard, Russell, and many others, see Moro (2017c).

such semantic notions as entity and property: all that is needed to define predication is the general compositional algorithm Merge, which forms SC by symmetrically combining the subject with the predicate.

All in all, our investigation on predication shows that a PredP is not a genuine expression of the cartographic research program, insofar as the postulation of a PredP inherently diminishes the explanatory power of this program. On the other hand, the reintroduction of exocentricity into the clausal spine permits unambiguously associating predication to a dedicated syntactic configuration, namely SC, and thus ultimately supports the cartographic research program. In this perspective, all sentences start with an exocentric predicative nucleus (SC) and are progressively and recursively expanded as endocentric phrases which are stacked in accordance with the detailed maps drawn in syntactic cartography.

References

- Adger, David (2013). *A syntax of substance*. Cambridge, MA: MIT Press.
- Belletti, Adriana (1990). *Generalized verb movement: Aspects of verb syntax*. Turin: Rosenberg & Sellier.
- Bowers, John (1993). 'The Syntax of Predication', *Linguistic Inquiry* 24 (4): 591-656.
- Bloomfield, Leonard (1933). *Language*. London: G. Allen & Unwin.
- Bussmann, Hadumod (1996). *Routledge Dictionary of Language and Linguistics*, translated and edited by Gregory Trauth and Kerstin Kazzazi. London: Routledge.
- Cecchetto, Carlo and Caterina Donati (2023). 'Labeling (Reduced) Structures: When VPs Are Sentences', *Linguistic Inquiry* 55 (1): 1-37.
- Chierchia, Gennaro and Sally McConnell-Ginet (2000). *Meaning and Grammar: An Introduction to Semantics*. Second Edition. Cambridge, MA: MIT Press.
- Chomsky, Noam (1956). 'Three models for the description of language', *I.R.E. Transactions on Information Theory* 2: 113-124.
- Chomsky, Noam (1957). *Syntactic structures*. The Hague: Mouton.
- Chomsky, Noam (1965). *Aspects of the theory of syntax*. Cambridge, MA: MIT Press.
- Chomsky, Noam (1981). *Lectures on Government and Binding: The Pisa Lectures*. Dordrecht: Foris.
- Chomsky, Noam (1986a). *Knowledge of language: Its nature, origin and use*. New York: Praeger.
- Chomsky, Noam (1986b). *Barriers*. Cambridge, MA: MIT Press.
- Chomsky, Noam (1988). *Language and Problems of Knowledge: The Managua Lectures*. Cambridge, MA: MIT Press.
- Chomsky, Noam (1993). *Lectures on Government and Binding: The Pisa Lectures*. Seventh edition. Berlin: de Gruyter.
- Chomsky, Noam (1995). *The Minimalist Program*. Cambridge, MA: MIT Press.
- Chomsky, Noam (2002). *Syntactic structures*. Second edition with an introduction by David W. Lightfoot. Berlin and New York: Mouton de Gruyter.
- Chomsky, Noam (2013). 'Problems of projection', *Lingua* 130: 33-49.
- Chomsky, Noam (2015). *Aspects of the theory of syntax*. 50th Anniversary Edition. Cambridge, MA: MIT Press.
- Chomsky, Noam (2019). *The UCLA Lectures*. Manuscript, Lingbuzz/005485.
- Cinque, Guglielmo (1990). *Types of $\bar{A}$ -dependencies*. Cambridge, MA: MIT Press.

- Cinque, Guglielmo (1999). *Adverbs and Functional Heads: A Cross-Linguistic Perspective*. Oxford: Oxford University Press.
- Cinque, Guglielmo (2010). *The Syntax of Adjectives: A Comparative Study*. Cambridge, MA: MIT Press.
- Cinque, Guglielmo, and Luigi Rizzi (2015). 'The Cartography of Syntactic Structures', in Bernd Heine and Heiko Narrog (eds.), *The Oxford Handbook of Linguistic Analysis*. Second Edition. Oxford: Oxford University Press, 65-78.
- Comrie, Bernard (1997). 'The Typology of Predicate Case Marking', in Joan L. Bybee, John Haiman, and Sandra A. Thompson (eds.), *Essays on Language Function and Language Type: Dedicated to T. Givón*. Amsterdam and Philadelphia: John Benjamins, 39-50.
- Cruschina, Silvio and Valentina Bianchi (2021). 'Mirative implicatures at the syntax-semantics interface: A surprising association and an unexpected move', in Andreas Trotzke and Xavier Villalba (eds.), *Expressive Meaning across Linguistic Levels and Frameworks*. Oxford: Oxford University Press, 86-107.
- den Dikken, Marcel (2006). *Relators and Linkers: The Syntax of Predication, Predicate Inversion, and Copulas*. Cambridge, MA: MIT Press.
- den Dikken, Marcel (2015). 'The morphosyntax of inalienably possessed noun phrases: The contribution of Hungarian', in Katalin É. Kiss, Balázs Surányi, and Éva Dékány (eds.), *Approaches to Hungarian: Volume 14: Papers from the 2013 Piliscsaba Conference*. Amsterdam and Philadelphia: John Benjamins, 121-145.
- Dowty, David R., Robert E. Wall, and Stanley Peters (1981). *Introduction to Montague Semantics*. Dordrecht: Reidel.
- Hale, Ken and Samuel Jay Keyser (2002). *Prolegomenon to a Theory of Argument Structure*. Cambridge, MA: MIT Press.
- Heim, Irene and Angelika Kratzer (1998). *Semantics in Generative Grammar*. Malden, MA and Oxford: Blackwell.
- Hoekstra, Teun and René Mulder (1990). 'Unergatives as copular verbs; locational and existential predication', *The Linguistic Review* 7(1): 1-80.
- Jespersen, Otto (1924). *The Philosophy of Grammar*. London: Allen & Unwin.
- Kayne, Richard S. (1994). *The antisymmetry of syntax*. Cambridge, MA: MIT Press.
- Kayne, Richard S. (2005). 'Some notes on comparative syntax, with special reference to English and French', in Guglielmo Cinque and Richard S. Kayne (eds.), *The Oxford Handbook of Comparative Syntax*. Oxford: Oxford University Press, 3-69.
- Larson, Richard K. (1988). 'On the Double Object Construction', *Linguistic Inquiry* 19(3): 335-391.
- Lepschy, Giulio C. (2000). *La linguistica del Novecento*. Bologna: Il Mulino.
- Longobardi, Giuseppe (1985). 'Su alcune proprietà della sintassi e della forma logica delle frasi copulari', in Annalisa Franchi De Bellis and Leonardo Maria Savoia (eds.), *Sintassi e morfologia della lingua italiana d'uso: Teorie e applicazioni descrittive*. Rome: Bulzoni, 211-213.
- Matthews, P.H. (2019). *What Graeco-Roman Grammar Was About*. Oxford: Oxford University Press.
- Mocci, Davide, and Tiziana Pontillo (2020). 'A Controversial Provision for the Nominative Ending: Nominal Sentences and Aṣṭādhyāyī 2.3.46', *Journal of the American Oriental Society* 140(1): 47-80.
- Mocci, Davide (2022). 'The possessive relation in Sanskrit bahuvrīhi compounds: Ellipsis or movement?', *Journal of South Asian Linguistics* 12(2): 1-30.

- Mocci, Davide (2024). 'The internal order of Sanskrit compounds: A dialogue between Pāṇini and generative grammar', in Savina Raynaud, Maria Paola Tenchini, and Enrica Galazzi (eds.), *History of Linguistics 2021: Selected papers from the 15th International Conference on the History of the Language Sciences (ICHoLS 15), Milan, 28 August – 1 September*. Amsterdam and Philadelphia: John Benjamins, 145-158.
- Montague, Richard (1973). 'The Proper Treatment of Quantification in Ordinary English', in K.J.J. Hintikka, J.M.E. Moravcsik, and Patrick Suppes (eds.), *Approaches to Natural Language*. Springer: Dordrecht.
- Moro, Andrea (1988). 'Per una teoria unificata delle frasi copulari', *Rivista di Grammatica Generativa* 13: 81-110.
- Moro, Andrea (1997a). *The raising of predicates: Predicative noun phrases and the theory of clause structure*. Cambridge: Cambridge University Press.
- Moro, Andrea (1997b). 'Dynamic antisymmetry: Movement as a symmetry-breaking phenomenon', *Studia Linguistica* 51: 50-76.
- Moro, Andrea (2000). *Dynamic Antisymmetry*. Cambridge, MA: MIT Press.
- Moro, Andrea (2011). 'Clause Structure Folding and the "Wh-in-Situ Effect"', *Linguistic Inquiry* 42(3): 389-411.
- Moro, Andrea (2013). *The Equilibrium of Human Syntax: Symmetries in the Brain*. New York: Routledge.
- Moro, Andrea (2017a). 'Copular sentences', in Martin Everaert and Henk van Riemsdijk (eds.), *The Wiley Blackwell Companion to Syntax*. Second Edition. Oxford: John Wiley & Sons Ltd, 1210-1232.
- Moro, Andrea (2017b). 'Existential sentences', in Martin Everaert and Henk van Riemsdijk (eds.), *The Wiley Blackwell Companion to Syntax*. Second Edition. Oxford: John Wiley & Sons Ltd, 1483-1508.
- Moro, Andrea (2017c). *A Brief History of the Verb to Be* (tr. by B. McClellan-Broussard). Cambridge, MA: MIT Press.
- Moro, Andrea (2019). 'The geometry of predication: A configurational derivation of the defining property of clause structure', *Philosophical Transactions of the Royal Society B* 375(1791): 1-7.
- Moro, Andrea (2023). 'Conjunctions in the left periphery: On symmetry-breaking in syntax', talk given at the 5th *International Workshop on Syntactic Cartography (IWSC2023)*, Beijing Language and Culture University, 29-30 October 2023, Beijing.
- Moro, Andrea (2024). *Parlo dunque sono. Istantanee sul linguaggio*. Nuova edizione accresciuta. Milan: Adelphi.
- Moro, Andrea and Ian Roberts (2024). 'The duality of syntax: Unstable structures, labelling and linearisation', *Natural Languages & Linguistic Theory* 42: 609-631.
- Napoli, Donna Jo (1989). *Predication theory: A case study for indexing theory*. Cambridge: Cambridge University Press.
- Partee, Barbara H., Alice ter Meulen, and Robert E. Wall (1990). *Mathematical Methods in Linguistics*. Dordrecht: Kluwer.
- Pereltsvaig, Asya (2007). *Copular Sentences in Russian: A Theory of Intra-Clausal Relations*. Dordrecht: Springer.
- Pollock, Jean-Yves (1989). 'Verb Movement, Universal Grammar, and the Structure of IP', *Linguistic Inquiry* 20(3): 365-424.
- Rizzi, Luigi (1982). *Issues in Italian Syntax*. Dordrecht: Foris.

- Rizzi, Luigi (1997). 'The Fine Structure of the Left Periphery', in Liliane Haegeman (ed.), *Elements of Grammar*. Dordrecht: Kluwer, 281-337.
- Rizzi, Luigi (2013). 'The functional structure of the sentence, and cartography', in Marcel den Dikken (ed.), *The Cambridge Handbook of Generative Syntax*. Cambridge: Cambridge University Press, 425-457.
- Rizzi, Luigi (2015). 'Notes on Labeling and Subject Positions', in Elisa di Domenico, Cornelia Hamann, and Simona Matteini (eds.), *Structures, Strategies and Beyond: Studies in Honour of Adriana Belletti*. Amsterdam and Philadelphia: John Benjamins, 17-46.
- Rizzi, Luigi (2016). 'Labeling, maximality and the head-phrase distinction', *The Linguistic Review* 33(1): 103-127.
- Rothstein, Susan Deborah (1983). *The syntactic forms of predication*. Doctoral Dissertation, Massachusetts Institute of Technology.
- Shlonsky, Ur and Giuliano Bocci (2019). 'Syntactic Cartography', *Oxford Research Encyclopedia of Linguistics*.
<https://oxfordre.com/linguistics/view/10.1093/acrefore/9780199384655.001.0001/acrefore-9780199384655-e-310>.
- Shlonsky, Ur and Luigi Rizzi (2018). 'Criteria Freezing in small clauses and the cartography of copular constructions', in Jutta Hartmann, Marion Jäger, Andreas Kehl, Andreas Konietzko and Susanne Winkler (eds.), *Freezing: Theoretical Approaches and Empirical Domains*. Berlin and Boston: De Gruyter Mouton, 29-65.
- Williams, Edwin (1980). 'Predication', *Linguistic Inquiry* 11(1): 203-238.
- Zwicky, Arnold M (1985). 'Heads', *Journal of Linguistics* 21(1): 1-29.